SELECT 2: A Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke

Statistical Analysis Protocol
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1.0.0

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List of abbreviations

ICA	Internal Carotid Artery
MCA	Middle Cerebral Artery
ASPECTS	Alberta Stroke Program Early CT Score
rCBF	Relative cerebral blood flow
CTP	CT Perfusion
MRP	Magnetic Resonance Perfusion
ADC	Apparent Diffusion Co-efficient
MRI	Magnetic Resonance Imaging
EVT	Endovascular Thrombectomy
MM	Medical Management only
DEFUSE3	Endovascular Therapy Following Imaging Evaluation for Ischemic Stroke 3
mRS	Modified Rankin Scale
ITT	Intention to treat
GLR	Generalized Likelihood Ratio
WMW	Wilcoxon-Mann-Whitney
ARD	average risk difference
CMH	Cochran-Mantel-Haenszel
NNT	number needed to treat
NIHSS	National Institutes of Health Stroke Scale
ICH	Intracranial Hemorrhage
sICH	Symptomatic Intracranial Hemorrhage
DSMB	Data Safety Monitoring Board

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1. Introduction

The SELECT2 trial is an investigator-initiated, phase III, international, multicenter, prospective, randomized, open-label, blinded-endpoint (PROBE) superiority study with adaptive enrichment two-arm parallel design, with 1:1 assignment to thrombectomy plus medical management (EVT arm) vs medical management only (MM arm). Randomization is conducted via a secure web-based interface. The subjects are followed for 1 year after the procedure.

2. Primary study aim

The primary aim of the study is to evaluate thrombectomy efficacy and safety in patients with acute ischemic stroke due to a large vessel occlusion in the anterior circulation (MCA M1 and ICA) and large ischemic core on either non-contrast CT (Alberta Stroke Program Early CT Score, ASPECTS: 3-5), CT or -advanced perfusion imaging (relative cerebral blood flow, [rCBF<30% lesion volume ≥50cc)] on CTP or diffusion-weighted MRI [(apparent diffusion co-efficient, ADC <620 x 10^{-6} mm^2/s lesion volume] on MRI: ≥50cc), treated within 0-24 hours from last known well.

3. Study Population

3.1 Inclusion criteria

1) Adults (18-85 years) with the final diagnosis of an acute ischemic stroke
2) NIHSS ≥ 6
3) Last known well to arterial puncture or medical management between 0 to 24 hours
4) Pre-stroke modified Rankin Scale score of 0-1
5) Eligible for thrombectomy or medical management
6) Signed Informed Consent obtained
7) Subject willing to comply with the protocol follow-up requirements
8) Anticipated life expectancy of at least 3 months

3.2 Exclusion criteria

1) Inability to undergo CT-Angiography and/or CT/MR Perfusion imaging (e.g., renal insufficiency, iodine/contrast allergy)
2) Co-morbid psychiatric or medical illnesses that would confound the neurological assessments
3) Treatment with IV thrombolytic therapy beyond 4.5 hours from last known well
4) Treated with IV thrombolytic therapy 3-4.5 hours after last known well AND any of the following:
    a. age >80
    b. current anticoagulant use
    c. history of both diabetes AND prior stroke
    d. NIHSS >25
    e. ischemic involvement of more than third of the MCA territory
5) Current participation in another investigational drug or device study.

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3.3 Specific Neuroimaging Inclusion Criteria

1) Proven large vessel occlusion in ICA or MCA-M1 occlusion (carotid occlusions can be cervical or intracranial, with or without tandem MCA lesions) determined by MRA or CTA
2) Large infarct-core lesion on at least one of the following:
    a. Non-Contrast CT (ASPECTS of 3-5),
    b. CT perfusion (rCBF<30% lesion ≥50cc),
    c. MRI-DWI (ADC<620 x 10^{-6} mm^2/s lesion ≥50cc),

3.3 Specific Neuroimaging Exclusion Criteria

1) Patients who have both ASPECTS of 6-10 on non-contrast CT AND ischemic core volume <50 cc on perfusion imaging
2) Patients with very large core on non-contrast CT i.e. ASPECTS ≤ 2.
3) Evidence of intracranial tumor (except small meningioma), acute intracranial hemorrhage, neoplasm, or arteriovenous malformation
4) Significant mass effect with midline shift
5) Evidence of internal carotid artery dissection that is flow limiting or aortic dissection
6) Intracranial stent implanted in the same vascular territory that precludes the safe deployment/removal of the neurothrombectomy device
7) Acute symptomatic arterial occlusions in more than one vascular territory confirmed on CTA/MRA (e.g., bilateral MCA occlusions, or an MCA and a basilar artery occlusion).
8) Signs of established infarct and large area of cerebral edema on non-contrast CT

4. Randomization

Randomization for the SELECT 2 trial takes place centrally using specifically designed randomization module in the online web-based REDCap trial database management system with the implementation of minimization with biased coin algorithm proposed by Pocock and Simon[1]. The randomization module is dynamic and ensures that the stratification of subgroups across the treatment arms is well-balanced, by balancing the treatment assignment along the marginal distribution of each stratification factors. The dynamic randomization system measures the imbalance of various stratification factors using predetermined weights at the onset of each randomization and assigns the treatment probability in such a way, that the group with the smallest imbalance receives the largest probability of assignment in the biased-coin randomization. Further details about the stratification factors and weights are provided in Appendix A. The randomization procedure was implemented UTH Biomedical Informatics group (UTH BIG) at University of Texas Health Science Center at Houston and maintained by Data management core at Center for Clinical and Translational Sciences at UT McGovern Medical School.

5. Baseline and follow-up assessments

All responsible investigators receive training in the systems for data collection and data entry and training in Good Clinical Practice (GCP). All investigators responsible for patient assessment participate in training in critical assessment items (NIHSS, mRS).

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The baseline assessments include NIHSS, pre-morbid mRS, demographic data, past medical history, previous stroke history and cerebral imaging information.

The treatment phase of the study is conducted in an open-label manner. All patients are followed up at 24hr, 5-7 days, 30 days, 90 days and 1 year unless death occurs. The assessment of functional outcomes is obtained by assessors blinded to the treatment arm and imaging parameters.

The trial management team are responsible for ensuring that all data are completed in a timely manner. Institutions receive small reimbursements for their staff time involved in data collection and in the provision of the intervention for this trial. Patients do not receive payment for participation.

6. Blinding

The treatment phase of the study is conducted in an open-label manner. The assessment of functional outcomes is obtained by assessors blinded to the treatment arm and imaging parameters. Only the Data and Safety Monitoring Board (DSMB) have access to interim data and results. The DSMB Co-Chairs are Dr. Lawrence Wechsler (University of Pennsylvania, US) and Dr. Stephen Davis (University of Melbourne, AU). Statisticians not involved in the DSMB will work on dummy datasets until the computer codes for statistical analysis are validated and will only have access to full data after the study enrollment, follow-up and data collection is completed and the dataset is locked.

7. Study design and sample size considerations

7.1 Adaptive enrichment design

The study implemented a novel adaptive enrichment design that not only allowed for the trial to potentially stop early for efficacy, but also allowed for selection and evaluation of the treatment effect in the most promising subpopulation if the futility rule was triggered at the interim (or final) analyses in the overall population. This adaptive design has been developed, well-studied, and implemented in the DEFUSE 3 trial[2], which assessed thrombectomy efficacy and safety in patients presenting within 6-16 hours from last known well. The details and theory of this adaptive enrichment design are summarized before. This adaptive enrichment design is an extension of standard group sequential design with an enrichment feature and is designed to test for efficacy or futility with two interim analyses and a final analysis[3,4]. This adaptive design was chosen because the preliminary data from the SELECT study suggested potential heterogeneity in thrombectomy treatment effect based on the different imaging profiles. The aim of this adaptive design is to reallocate future accrual and study treatment effects only in the most promising subpopulation if there was evidence that EVT was futile in the overall population at interim (or final) analyses. By limiting the subsequent enrollment only in the most promising subpopulation, a larger number of patients would eventually be enrolled in this subgroup for efficacy testing compared to traditional designs.

This design uses the Generalized Likelihood Ratio (GLR) statistic (Kullback-Leibler criterion) as criteria to identify the subgroup, which has the best chance of showing a benefit from EVT. Through these criteria, it can achieve optimal balance simultaneously between the estimated size of the treatment effect and the sample size of the subgroup. By considering the CT ASPECTS score, CTP core volume and treatment

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time window, the patients are split into six disjoint groups, that is Group 1 (ASPECTS 3-5 and CTP/MRP core volume <50cc) at early time window 0-12 hours (denoted as **G1A**), Group 1 at late time window 12-24 hours (denoted as **G1B**), Group 2 (ASPECTS 6-10 and CTP/MRP core volume ≥50cc) at early time window (denoted as **G2A**), Group 2 at late time window (denoted as **G2B**), Group 3 (ASPECTS 3-5 and CTP/MRP core volume ≥50cc) at early time window (denoted as **G3A**), and Group 3 at late time window (denoted as **G3B**).

Considering a prior assumption that the EVT effect is largest in the patients with good imaging profile and treated within the shortest time window, we assume that the highest effect of EVT will be observed in G1A. Then the effects get decreased gradually in G1B, G2A, G2B, G3A, and the lowest effect will be observed in G3B. According to this adaptive design, when a most promising subgroup needs to be selected at the time of interim (or final) analysis, we will compare the GLR statistic among five cumulative groups, that is, G1A, G1A+G1B (i.e., Group 1), G1A+G1B+G2A, G1A+G1B+G2A+G2B (i.e., Group 1+Group 2), and G1A+G1B+G2A+G2B+G3A.

7.2 Interim and final analyses

The study design incorporated two interim analyses after enrollment and completion of 90-day follow-up of 200 and 380 patients to assess the efficacy of thrombectomy vs medical management in patients with large core on one or both imaging modalities using primary outcome – distribution of mRS score at 90-day follow-up. At each analysis, an efficacy boundary was set to decide if the study should be stopped for efficacy while controlling the overall (one-sided) Type I error rate at 2.5%. At each interim analysis, a futility boundary was set to decide if the study should continue recruitment in the overall group, shift accrual and testing to a subgroup, or stop the trial entirely. If the futility boundary was crossed and the optimal subgroup was to be selected, the maximum analyzed sample size would not have been fixed but a random variable (smaller than the fixed maximum sample size 560). Due to the reduction in the maximum number of patients available for analysis at the end of the study, our design allowed for an easier futility stop after subgroup selection. The details regarding efficacy and futility boundaries is provided in Appendix B.

At first interim analysis (enrollment and completion of 90-day follow-up for first 200 patients), the null hypothesis is tested in the study cohort:

1. If neither efficacy nor futility boundary is crossed, the trial continues enrollment to the next interim/final analysis.
2. If the efficacy boundary is crossed, the trial stops and efficacy is declared in the study cohort.
3. If the futility boundary is crossed, the optimal subgroup is selected based on the GLR statistic and the null hypothesis is tested in this selected subgroup:
    a. If neither boundary is crossed, the trial will continue with enrollment limited to the selected subgroup

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b. If the efficacy boundary is crossed, the trial stops and efficacy is declared in the selected subgroup.

c. If the futility boundary is crossed, the trial stops for futility.

After the first interim analysis, the trial proceeds to this stage under two scenarios: (1) continuing enrollment in the overall population (i.e., option 1 above) and (2) limiting enrollment to a selected subgroup (i.e., option 3a above). When scenario (1) happens, the testing at the 2nd interim analysis at enrollment and completion of 90-day follow-up of 380 patients is identical to the first interim analysis. When scenario (2) happens, the null hypothesis is tested in that selected subgroup:

1. If neither boundary is crossed, the trial continues to the final analysis with an enrollment of 560-380=180 additional patients limited to the selected subgroup.
2. If the efficacy boundary is crossed, the trial stops and efficacy is declared in the selected subgroup.
3. If the futility boundary is crossed, the trial stops for futility.

After the second interim analysis, the trial proceeds to final stage under two scenarios: (1) continuing enrollment in the overall population and 2) limiting enrollment to a selected subgroup. After enrollment and completion of 90-day follow-up of further 180 patients, the null hypothesis is tested in either 1) overall population or 2) selected subgroup.

1. If the efficacy boundary is crossed, EVT is declared efficacious.
2. If the efficacy boundary is not crossed, EVT is declared as not superior to standard medical management.

7.3 Sample size estimation

Power calculations and sample size estimation was undertaken based on the rationale provided in by Lai, Lavori and Liao (2014)[3] and completed using R package ASSISTant by the Dr. Chunyan Cai, PhD of Center for Clinical & Translational Sciences, Dept. of Internal Medicine, UT McGovern Medical School. Based on prior reported data from SELECT study[5] as well as expert assessment on treatment effect by taking into consideration the non-randomized design of SELECT and potential selection biases, the following distributions were hypothesized for the mRS at 90 days in EVT and MM groups for SELECT 2 in the overall population.

	mRS at day 90							
	0	1	2	3	4	5	6	Total
Endovascular Thrombectomy	4%	8%	12%	16%	19%	13%	28%	100%
Medical Management	1%	4%	9%	14%	19%	10%	43%	100%

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This distribution corresponds to a standardized difference of 0.34 for the primary analysis. Assuming the above effect size, the fixed sample size for a non-adaptive design requires a total of 376 patients (188/arm) to have 90% power at a one-sided alpha of 2.5% (Wilcoxon-Mann-Whitney test). Extra 184 patients for the adaptive design were added to reach a maximum sample size of 560. The size of this increase was selected based on simulations and allows preservation of the desired operating characteristics while allowing potential shrinkage in standardized difference in mRS score distribution to as low as 0.30 with reasonable statistical power, since the above effect estimate may be optimistic. This approach has been validated and utilized for DEFUSE3 trial before[2]. Two interim analyses were pre-planned to be conducted based on the outcomes of 200 and 380 patients respectively. Through extensive simulation studies, it was shown that with 560 patients in total for the adaptive design, statistical power of 91% will be achieved to test for the hypothesized standardized difference of 0.34 in the overall population (assuming homogeneous EVT effect across six disjoint groups described in section 1.1) by controlling an overall one-sided Type I error rate at 2.5%.

7.4 Operating characteristics of the adaptive design

Simulations (5000 simulated trials) were conducted to evaluate the performance of the adaptive design (max n=560) and compare it to a traditional fixed sample-size design (fixed n=560) under various scenarios (see the illustrative table below). For the simulations the effect size is expressed as a standardized effect with normal approximation, where a standardized difference of 0.32 or 0.30 corresponds to a conservative projected effect of EVT (anticipated standardized difference 0.34; see section of Power and sample size considerations). The assumptions were that the observed proportions of patients in G1A, G1B, G2A, G2B, G3A, G3B were 14%, 6%, 14%, 6%, 42%, and 18%, respectively.

Scenario	Standardized difference in groups						Average standardized difference	Adaptive Design		Fixed Design	
	G1A	G1B	G2A	G2B	G3A	G3B		Average No. randomize	Power	Number randomized	Power
#0	0	0	0	0	0	0	0	426	2.0%	560	2.5%
#1	0.32	0.32	0.32	0.32	0.32	0.32	0.32	376	86.0%	560	95.8%
#2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	393	81.1%	560	93.3%
#3	0.4	0.4	0.4	0.4	0	0	0.16	500	74.1%	560	45.7%
#4	0.4	0.4	0.4	0.4	-0.1	-0.1	0.1	502	78.5%	560	51.2%
#5	0.4	0.4	0.3	0.3	0	-0.1	0.122	513	67.1%	560	29.2%
#6	0.5	0.5	0	0	0	0	0.1	503	70.6%	560	21.2%
#7	0.5	0	0	0	0	0	0.07	503	60.5%	560	12.8%

Under the null (Scenario #0), the adaptive design controls the total Type 1 error below 2.5%, stops early for futility 59.1% of the time, and the average number of randomizations is 426. If the standardized difference is uniform across cells at 0.32 (scenario #1), the fixed-sample design is optimal, but the adaptive design results in only a small loss of power (from 95.8% to 86.0%),but reduced the expected sample size (from 560 to 376). We observed similar findings when the standardized

difference is uniform across cells at 0.3 (scenario #2). The adaptive design performs much better (higher power and smaller expected sample size) than the fixed-sample design when the distribution ofstandardized difference across the subgroups is in accord with the biological assumptions (scenarios #3, #4 and #5). If the standardized difference is concentrated only in Group 1 (scenario #6) or Group 1A (scenario #7), the adaptive design maintains power (70.6% in scenario #6 and 60.5% in scenario #7) while the conventional design collapses (21.2% power in scenario #6 and 12.8% in scenario #7).

7.5 DSMB review after enrollment and completion of 90-day follow-up of 300 patients

First interim analysis after enrollment and study completion of 200 patients was conducted on February 2nd, 2022. The data safety monitoring board reviewed the results and recommended continuation of the study.

Following the external signal constituted by the presentation and publication of the results of RESCUE JAPAN Limit[6], the data safety monitoring board requested to review the efficacy and safety outcomes after enrollment and completion of 90-day follow-up of 300 patients. As the original study design included pre-specified interim analyses and specified efficacy and futility boundaries at 200 and 380 patients as detailed in Appendix B, the SELECT2 steering committee met to discuss how to address the DSMB request. At the meeting, the decision was made to provide the DSMB with the requested data but without making any alteration to the study design or incorporating any additional interim analyses.

The DSMB met on September 9th, 2022 and reviewed the efficacy and safety data for first 300 patients who completed the study. They also reviewed available outcomes (30d and 90d follow-up) for all enrolled patients until September 1, 2022. Based on the review, using pre-specified boundaries for interim analyses and after examination of the data from 300 pts completing 90-day follow-up, the DSMB recommended that the SELECT 2 trial should immediately stop any further enrollment as the boundary for efficacy has been crossed in favor of EVT.

Based on this recommendation, SELECT 2 steering committee stopped further recruitment of participants and rejected null hypothesis of no difference in treatment effect between study arms at the family-wise Type I error rate of 0.05.

8. Considerations for data collection and parameters definition

- Functional outcomes at 30d and 90d follow-up will be collected by assessors blinded to treatment allocation and imaging characteristics
- CT ASPECTS, occlusion location, ischemic core volume, degree of reperfusion on post-procedure angiogram, infarct volume and infarct growth measures will be centrally adjudicated by imaging core lab
- Perfusion imaging will be reprocessed centrally if raw images are available
- In cases with suboptimal perfusion imaging acquisition, core volume estimates will be determined by imaging core lab evaluation of CT hypodensity core volume.
- Hemorrhage categories, neurological worsening and procedural complications will be centrally adjudicated by the imaging core lab and independent clinical monitor.

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- Symptomatic hemorrhage will be adjudicated centrally by core lab imaging review and independent clinical monitor

9. Definitions of outcomes

9.1 Primary outcome

- The 90-day functional outcome, as measured by the modified Rankin Scale score distribution.

9.2 Secondary outcomes

- Functional independence, defined as mRS score of 0-2 at 90-day follow-up
- Independent ambulation, defined as mRS score of 0-3 at 90-day follow-up
- Symptomatic intracranial hemorrhage (sICH) as measured by the SITS-MOST criteria (local PH2/remote PH hemorrhagic transformation) associated with a ≥4-point increase on the NIHSS score or death within 24 hours
- Neurological worsening, defined as defined as a ≥4-point increase on the NIHSS score due to the stroke itself within 24 hours
- Mortality within 90-day follow-up
- Procedural complications, defined as cumulative incidence of
    - Surgical site hematoma
    - Surgical site infection
    - Vascular injury (peripheral or intracranial dissection, perforation or others) caused by the endovascular procedure
- Successful reperfusion in the EVT group, defined as modified Thrombolysis in Cerebral Ischemia (mTICI) grade of 2b or higher
- Discharge location
- Early neurological improvement, defined as improvement of ≥8 points on NIHSS at 24 hours of presentation or an NIHSS of 0-1
- Quality of life score, as measured using NeuroQOL at 90-day follow-up
- The 1-year functional outcome, as measured by the modified Rankin Scale score*
- The rate of functional independence, defined as mRS score of 0-2 at 1-year follow-up*
- The rate of independent ambulation, defined as mRS score of 0-3 at 1-year follow-up*
- Quality of life score, as measured using NeuroQOL at 1 year follow-up*

*These outcomes will not be available at the time of primary manuscript and will be reported later in separate manuscripts.

9.3 Tertiary outcomes*

- Infarct volume on MRI DWI sequence (or CT if MRI not feasible) 24 hours to 7 days after randomization
- Lesion growth between the ischemic core on baseline imaging and the follow-up infarct volume
- Parenchymal Hemorrhage type 2

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- Subarachnoid hemorrhage
- Hemicraniectomy during index hospitalization
- Length of hospital stay

*Some or all of these outcomes may be reported in publications subsequent to primary paper

10. Statistical analysis

10.1 Analysis principles and general considerations

- The analyses are based on the total sample of 352 participants recruited prior to the study stopping as described in Section 7.5.
- All outcomes and analyses are prospectively categorized as primary, secondary, tertiary or exploratory.
- No formal adjustments will be undertaken to constrain the overall type I error associated with the secondary or tertiary outcomes and exploratory analyses. Their purpose is to supplement evidence from the primary analysis to fully characterize the treatment effect. Results from the secondary and tertiary analyses will be interpreted in this context. All confidence intervals will be reported at 95% level of confidence.
- Analysis for the primary outcome, and all secondary efficacy and safety outcomes analyses will be conducted on an intention-to-treat (ITT) basis so that all patients will be analyzed in the group to which they were randomized irrespective of whether they received the allocated treatment. The primary analysis strategy based on ITT is defined as follows:
    - Is based on an ITT design that aims to collect all outcome data on all randomized subjects;
    - Includes a main analysis that keeps subjects in their randomized groups, analyzes all available outcome data, and is valid under a named plausible assumption about the missing data;
    - Includes tipping point analysis[7] (that consider a range of alternative values for missing outcomes data to evaluate the extreme distribution of missing data for primary outcome where the results change) as an alternative to missing at random assumption
    - All individuals are included in sensitivity analyses
- Patients with mRS scores of 6 (death) will be merged with patients who achieved scores of 5 (bedridden with constant care required) to avoid counting a shift from mRS 6 to 5 as a treatment success[9, 10].
- Key analyses will be repeated in the subpopulations including 1) As-treated population and 2) per-protocol population. Further details regarding these subpopulations are provided in section 11.
- Subgroup analyses will be carried out, irrespective of whether there is a significant treatment effect on the outcome. Their purpose is to supplement evidence from the primary analysis to help to fully characterize the treatment effect. Results from subgroup analyses will be

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interpreted in this context. Further details regarding subgroup analyses are provide in section 11.

- The data from all investigative sites will be pooled based on the assumption of comparable clinical aspects such as the common protocol used, adequate monitoring on protocol compliance, and the centralized data gathering and validation systems across all study sites. To assess poolability, a subgroup analysis based on location (US vs non-US sites) will also be included. Additional details regarding this analysis are provided in section 10.3.4.
- Analyses will be conducted primarily using Stata and R statistical software.

10.2 Trial reporting

10.2.1 Trial profile

Flow of patients through the study will be displayed in a standard CONSORT diagram (Supplementary Figure 1 in appendix). The report will include the number of patients included, withdrawn, lost to follow up, the number who received the allocated treatment and were analyzed.

10.2.2 Patient characteristics and baseline comparisons

To assess balance, description of the specified baseline characteristics will be presented for the two treatment arms (Table 1 – see Appendix). Discrete variables will be summarized as frequencies and percentages. Unless otherwise indicated in the tables, percentages will be calculated according to the number of patients for whom data are available. If there are more than 5% missing values, the denominator will be added in a footnote in the corresponding summary table. Continuous variables will be summarized by use of either mean and standard deviation (SD) or median and interquartile range (IQR). Durations and time intervals will be summarized by medians and IQRs.

10.2.3 Outcome reporting

The description of the specified primary and secondary clinical outcomes will be presented for the two treatment arms (Table 2 – see Appendix).

10.2.3.1 Primary outcome - The 90-day functional outcome, as measured by the modified Rankin Scale score

10.2.3.1.1 Outcome measure

The distribution of scores on the modified Rankin Scale (mRS) at day 90. Patients with mRS scores of 6 (death) will be merged with patients who achieved scores of 5 (bedridden with constant care required) to avoid counting a shift from mRS 6 to 5 as a treatment success as they are assigned same utility scores[9, 10].

10.2.3.1.2 Statistical hypothesis

The null hypothesis (to be rejected) is that the distribution of scores on the modified Rankin Scale (mRS) at day 90 in patients assigned to EVT arm does not differ from the distribution of scores on the modified Rankin Scale (mRS) at day 90 in patients assigned to MM arm.

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10.2.3.1.3 Treatment of missing values

All efforts are put forth to ensure complete follow-up. If the primary outcome (mRS at 90 days) cannot be assessed in the clinic, it will instead be obtained by phone using a structured interview. Based on prior stroke trials data, expected missingness in primary outcome is very low. Nevertheless, a small amount of missing data may be inevitable. All randomized subjects are to be included in the primary ITT outcome analysis so there is a need for a consistent ITT-based strategy of dealing with any missing primary outcome values.

Following clinical input from, and discussions with, the SELECT2 Steering Committee, the primary outcome is assumed to be missing-at-random (MAR), i.e. the missingness of the data may depend on the observed data, but not on the missing data. Important explanatory and auxiliary variables are collected and will be examined to assess the plausibility of the MAR assumption. We will perform multiple imputations for missing data where data will be imputed 30 times by selecting random values from a distribution of potential values based on strata (values) of all observed explanatory and auxiliary variables. Specifically, we will build an ordinal logistic regression model including explanatory and auxiliary variables to impute modified Rankin Scale scores for missing observations. Estimates will be pooled across the imputed datasets in accordance with Rubin's rule[8].

Tipping point analysis[7] (that consider a range of alternative values for missing outcomes data to evaluate the extreme distribution of missing data for primary outcome where the primary results change) will be performed as an alternative analysis not relying on missing at random assumption.

10.2.3.1.4 Analysis method

For primary outcome (distribution of 90-day mRS score), the treatment effect will be expressed as

- The Wilcoxon-Mann-Whitney measure of superiority (WMW), with its 95% confidence interval and p-value, merging mRS categories 5 and 6 to avoid transition from mRS 6 to 5 counting as treatment success as they are assigned same utility scores [9, 10].

For comparability with other studies, the following transformations of the WMW will also be reported:

- The average risk difference (ARD), defined as the probability that a randomly selected subject from the endovascular group has a better functional outcome than a randomly selected subject from the control group, with its 95% confidence interval, where ARD = 2(WMW – 0.5).
- The average number needed to treat for benefit (NNT), with its 95% confidence interval, where NNT = 1 / ARD.
- The Wilcoxon-Mann-Whitney Generalized Odds with its 95% confidence interval, merging mRS categories 5 and 6 to avoid transition from mRS 6 to 5 counting as treatment success as they are assigned same utility scores [9, 10].

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10.2.3.2 Secondary outcome - functional independence, defined as mRS score of 0-2 at 90-day follow-up

10.2.3.2.1 Outcome measure

The proportion of patients exhibiting a modified Rankin Scale (mRS) score of 0-2 at day 90.

10.2.3.2.2 Statistical hypothesis

The null hypothesis (to be rejected) is that the proportion of patients exhibiting a modified Rankin Scale (mRS) score of 0-2 at day 90 does not differ significantly between patients assigned to EVT arm and patients assigned to MM arm.

10.2.3.2.3 Treatment of missing values

As described in section 10.2.3.1.3, these data (modified Rankin Scale scores at 90-day follow-up) will be imputed using multiple imputations. The values for missing mRS 0-2 will be extrapolated from the imputed 90-day mRS distribution and used to obtain estimates of effect, which then will be pooled across imputed datasets using Rubin's rule.

10.2.3.2.4 Analysis method

The treatment effect will be expressed as

- The proportion of patients exhibiting a modified Rankin Scale (mRS) score of 0-2 at day 90
- The relative risk with its 95% confidence interval and p-value, calculated using modified Poisson regression

10.2.3.3 Secondary outcome - independent ambulation, defined as mRS score of 0-3 at 90-day follow-up

10.2.3.3.1 Outcome measure

The proportion of patients exhibiting a modified Rankin Scale (mRS) score of 0-3 at day 90.

10.2.3.3.2 Statistical hypothesis

The null hypothesis (to be rejected) is that the proportion of patients exhibiting a modified Rankin Scale (mRS) score of 0-3 at day 90 does not differ significantly between patients assigned to EVT arm and patients assigned to MM arm.

10.2.3.3.3 Treatment of missing values

Treatment of missing values will be performed similar to secondary outcome mRS 0-2 as described in section 10.2.3.2.3.

10.2.3.3.4 Analysis method

The treatment effect will be expressed as

- The proportion of patients exhibiting a modified Rankin Scale (mRS) score of 0-3 at day 90
- The relative risk with its 95% confidence interval and p-value, calculated using modified Poisson regression

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10.2.3.4 Secondary outcome - symptomatic intracranial hemorrhage (sICH) as measured by the SITS-MOST criteria

10.2.3.4.1 Outcome measure

The proportion of patients exhibiting symptomatic intracranial hemorrhage (sICH) as measured by the SITS-MOST criteria within 24 hours.

10.2.3.4.2 Statistical hypothesis

The null hypothesis (to be rejected) is that the proportion of patients exhibiting symptomatic intracranial hemorrhage (sICH) are similar between patients assigned to EVT arm and patients assigned to MM arm.

10.2.3.4.3 Treatment of missing values

Symptomatic ICH is assumed to be missing-at-random (MAR), i.e. the missingness of the data may depend on the observed data, but not on the missing data. Important explanatory and auxiliary variables are collected and will be examined to assess the plausibility of the MAR assumption. We will perform multiple imputations for missing data where data will be imputed 30 times by selecting random values from a distribution of potential values based on strata (values) of all observed explanatory and auxiliary variables. Specifically, we will build a logistic regression model including explanatory and auxiliary variables to impute symptomatic ICH for missing observations. Estimates will be pooled across the imputed datasets in accordance with Rubin's rule[8].

10.2.3.4.4 Analysis method

The treatment effect will be expressed as

- The proportion of patients exhibiting symptomatic intracranial hemorrhage (sICH)
- The relative risk with its 95% confidence interval and p-value, calculated using modified Poisson regression

10.2.3.5 Secondary outcome – neurological worsening, defined as defined as a ≥4-point increase on the NIHSS score within 24 hours due to the stroke itself

10.2.3.5.1 Outcome measure

The proportion of patients exhibiting neurological worsening, defined as a ≥4-point increase on the NIHSS score within 24 hours due to the stroke itself.

10.2.3.5.2 Statistical hypothesis

The null hypothesis (to be rejected) is that the proportion of patients exhibiting neurological worsening are similar between patients assigned to EVT arm and patients assigned to MM arm.

10.2.3.5.3 Treatment of missing values

Treatment of missing values will be performed similar to secondary outcome symptomatic ICH as described in section 10.2.3.4.3.

10.2.3.5.4 Analysis method

The treatment effect will be expressed as

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- The proportion of patients exhibiting neurological worsening
- The relative risk with its 95% confidence interval and p-value, calculated using modified Poisson regression

10.2.3.6 Secondary outcome – all-cause mortality within 90-day follow-up

10.2.3.6.1 Outcome measure

The proportion of patients exhibiting all-cause mortality within 90-day follow-up.

10.2.3.6.2 Statistical hypothesis

The null hypothesis (to be rejected) is that the proportion of patients exhibiting all-cause mortality are similar between patients assigned to EVT arm and patients assigned to MM arm.

10.2.3.6.3 Treatment of missing values

Treatment of missing values will be performed similar to secondary outcome mRS 0-2 as described in section 10.2.3.2.3.

10.2.3.6.4 Analysis method

The treatment effect will be expressed as

- The proportion of patients exhibiting mortality
- The relative risk with its 95% confidence interval and p-value, calculated using modified Poisson regression

10.2.3.7 Secondary outcome – Procedural complications

10.2.3.7.1 Outcome measure

The proportion of patients demonstrating surgical site complications, defined as a cumulative measure of surgical site hematoma, surgical site infection and vascular injury (peripheral or intracranial dissection, perforation or other) caused by endovascular procedure.

10.2.3.7.2 Statistical hypothesis

No hypothesis will be evaluated.

10.2.3.7.3 Treatment of missing values

Missing data for surgical site complications will not be imputed.

10.2.3.7.4 Analysis method

The count and proportion of patients exhibiting procedural complications will be reported

10.2.3.8 Secondary outcome – Successful reperfusion, defined as modified Thrombolysis in Cerebral Ischemia (mTICI) grade of 2b or higher

10.2.3.8.1 Outcome measure

The proportion of patients with Successful reperfusion, defined as modified Thrombolysis in Cerebral Ischemia (mTICI) grade of 2b or higher.

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10.2.3.8.2 Statistical hypothesis

No hypothesis will be evaluated.

10.2.3.8.3 Treatment of missing values

Missing data for successful reperfusion will not be imputed.

10.2.3.8.4 Analysis method

The count and proportion of patients achieving successful reperfusion will be reported

10.2.3.9 Secondary outcome – Discharge location

10.2.3.9.1 Outcome measure

The proportion of patients receiving discharge to home, other acute care facility, inpatient rehabilitation facility, skilled nursing facility, hospice care and in-hospital deaths.

10.2.3.9.2 Statistical hypothesis

The null hypothesis (to be rejected) is that the proportion of discharge locations does not differ significantly between patients assigned to EVT arm and patients assigned to MM arm.

10.2.3.9.3 Treatment of missing values

Missing data for discharge location will not be imputed

10.2.3.9.4 Analysis method

The treatment effect will be expressed as

- The proportion (%) of patients receiving discharge to
    1) home,
    2) other acute care facility,
    3) inpatient rehabilitation facility,
    4) skilled nursing facility,
    5) hospice care and
    6) in-hospital deaths

10.2.3.10 Secondary outcome – Early Neurological Improvement

10.2.3.10.1 Outcome measure

The proportion of patients exhibiting early neurological improvement, defined as improvement of ≥8 points on NIHSS at 24 hours of presentation or an NIHSS of 0-1.

10.2.3.10.2 Statistical hypothesis

The null hypothesis (to be rejected) is that the proportion of patients exhibiting early neurological improvement between patients assigned to EVT arm and patients assigned to MM arm.

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10.2.3.10.3 Treatment of missing values

Treatment of missing values will be performed similar to secondary outcome symptomatic ICH as described in section 10.2.3.4.3.

10.2.3.10.4 Analysis method

The treatment effect will be expressed as

- The proportion of patients exhibiting early neurological improvement
- The relative risk with its 95% confidence interval and p-value, calculated using modified Poisson regression

10.2.3.11 Secondary outcome – Quality of life score at 90-day follow-up

10.2.3.11.1 Outcome measure

The distribution of the quality of life score, measured using NeuroQoL.

10.2.3.11.2 Statistical hypothesis

The null hypothesis (to be rejected) is that the distribution of the quality of life score does not differ significantly between patients assigned to EVT arm and patients assigned to MM arm.

10.2.3.11.3 Treatment of missing values

Missing data for length of the quality of life score will not be imputed.

10.2.3.11.4 Analysis method

The treatment effect will be expressed as

- The distribution of quality of life scores, using median (IQR)
- The difference in median values with its 95% confidence interval and p-value, obtained using median regression

10.2.4 Subgroup analysis

The primary efficacy analysis as well analysis for functional independence (mRS 0-2) and independent ambulation (mRS 0-3) will be repeated in following subgroups as pre-specified subgroup analysis. Effect size and 95% CI will be provided in a forest-plot figure (Figure 3).

Subgroup	**Strata**
Age	**<70, ≥70**
Presentation NIHSS	**<20, ≥20**
Clot location	**ICA, MCA-M1**
Time from Last known well to randomization	**0-12, 12-24** **0-6, 6-24**

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Perfusion (rCBF <30%) lesion volume	**<70ml, ≥70ml** **<100ml, ≥100ml** **<150ml, ≥150ml**
CT ASPECTS	**6-10, 3-5, 0-2 (based on core lab ASPECTS)**
Perfusion mismatch profile (mismatch ratio ≥ 1.8 & mismatch volume ≥ 15 ml) present	**yes, no**
Perfusion mismatch profile (mismatch ratio ≥ 1.2 & mismatch volume ≥ 10 ml) present	**yes, no**
Trial population excluding a) ASPECTS 6-10 within 6 hours of last known well AND b) ASPECTS 6-10 with perfusion (rCBF <30%) lesion volume <70 ml within 6-16 hours of last known well	**yes, no**
Trial population with CT ASPECTS <6 and perfusion (rCBF<30%) lesion volume ≥70	**yes, no**
Hemisphere	**right, left**
Geographic location	**US sites, non-US sites**

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10.2.5 Tables and figures for the main manuscript

Table 1 will report key baseline characteristics of participants by patient group. Table 2 will report the primary and key secondary outcomes. Figure 1 will report the "Grotta bar" i.e. stacked bar chart of each grade on the mRS in each treatment group assignment in intention-to-treat analysis. Figure 2 will report the "Grotta bar" i.e. stacked bar chart of each grade on the mRS in each treatment group assignment in pre-specified populations including a) As-treated population and b) Per-protocol population. Figure 3 will be the forest plot demonstrating effect size with 95% CI for primary outcome in selected subgroups described in section 10.2.4. CONSORT diagram will be presented as supplementary figure 1.

11. Pre-specified subpopulation analyses

11.1 As-treated population

Due to the nature of the study, some subjects in the endovascular treatment plus medical management group may not receive endovascular therapy. One instance when this may occur is if symptoms resolve

spontaneously between randomization and start of the endovascular procedure. Another potential reason is the absence of an arterial occlusion on the baseline angiogram (due to spontaneous or thrombolysis-induced recanalization). Although it would be rare, we may also see some subjects who are randomized to the medical management only group, but are treated with endovascular therapy. Therefore, after completion of the study, we will repeat the primary analysis (the Wilcoxon-Mann-Whitney measure of superiority) within the adaptive design sample, but with patients categorized under the as-treated principle.

11.2 Per-protocol population

Due to the nature of the study, some subjects in the study may have had significant protocol violation, including eligibility criteria and treatment received. A per-protocol analysis including only individuals who meet all eligibility criteria, including imaging evaluation as determined by the central core lab and received treatment as indicated during randomization will also be performed.

12. Pre-specified exploratory analyses

Following pre-specified tertiary analyses will be undertaken in patients who received EVT or best medical management. These analyses may or may not be reported in the primary trial manuscript.

- Evaluate the association between imaging profiles with thrombectomy clinical outcomes and treatment effect
- Thrombectomy efficacy analysis by definitions of large core by various imaging modalities
    - ASPECTS 5 or less, regardless of ischemic core size
    - Ischemic core ≥ 50mL regardless of CT ASPECTS
    - ASPECTS 5 or less & ischemic core ≥ 50mL
- Functional independence by imaging modalities
- Heterogeneity of treatment effect by imaging modality
- Assessment of ceiling effect of ischemic core volume on thrombectomy outcomes
- Assessment of ceiling effect of CT ASPECTS on thrombectomy outcomes
- Thrombectomy effect modification by CT ASPECTS and perfusion (rCBF <30%) lesion volume together
- Impact of presence of mismatch on thrombectomy outcomes using 1) mismatch ratio ≥1.8 & mismatch volume ≥15ml and 2) mismatch ratio ≥1.2 & mismatch volume ≥10ml
- Overestimation of infarct core on baseline perfusion imaging, as processed by core lab
- Effect of total ischemic lesion volume [determined as a union of hypodensity volume on NCCT (manually delineated by core lab neurologist) and reprocessed CT perfusion rCBF<30% tissue volume if technically feasible] on efficacy and safety of endovascular thrombectomy. In cases where obtaining a union of two volumes is not feasible, the larger of the manually delineated CT hypodensity volume and reprocessed rCBF<30% tissue volume will be considered.
- Effect modification of Age on the association between endovascular thrombectomy and clinical outcomes

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- Effect modification of stroke severity on the association between endovascular thrombectomy and clinical outcomes
- Effect modification of Occlusion locations on the association between endovascular thrombectomy and clinical outcomes
- Effect modification of time to Treatment on the association between endovascular thrombectomy and clinical outcomes
- Effect modification of Ischemic core volume on the association between endovascular thrombectomy and clinical outcomes
- Effect modification of ASPECTS score on the association between endovascular thrombectomy and clinical outcomes
- Effect modification of Presence of mismatch on the association between endovascular thrombectomy and clinical outcomes
- Effect modification due to left vs right hemisphere involvement on the association between endovascular thrombectomy and clinical outcomes
- Effect modification of Transfer status on the association between endovascular thrombectomy and clinical outcomes
- Evaluation of workflow
- Time metrics
- Cost-effectiveness of endovascular thrombectomy procedure
- Adjunctive benefit of thrombolysis
- Presence of tandem occlusions
- Tenecteplase vs Alteplase use
- Impact of outcome evaluation using utility-weighted modified Rankin Scale score
- Discharge location
- Time to DNR-CC decision for patients transitioned to comfort care

Following pre-specified tertiary analyses will be undertaken in patients who received EVT.

- Anesthesia technique used
- Reperfusion status after thrombectomy
- Interventional parameters
- Procedural metrics

13. Acknowledgement

We would like to thank Dr. Chunyan Cai, PhD at Center for Clinical and Translational Science, Dept. of Internal Medicine, UT McGovern Medical School for her role in 1) developing and implementing the randomization module and 2) performing power calculations and operating characteristics analysis.

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15. Illustrative tables and figures:

Table 1. Baseline characteristics of intention-to-treat population stratified based on allocation of treatment.

		Total	**Allocated to endovascular thrombectomy + medical management (EVT arm)**	**Allocated to medical management only (MM arm)**
		N=xx	N=xx	N=xx
Age (years), median [IQR]				
Sex, n(%)	Males			
	Females			
H/o Hypertension				
H/o Congestive Heart Failure, n(%)				
H/o Coronary Artery Disease, n(%)				
H/o Atrial Fibrillation, n(%)				
H/o Diabetes Mellitus, n(%)				
H/o Transient Ischemic Attack, n(%)				
H/o Stroke, n(%)				
Hemisphere affected, n(%)	Right			
	Left			
Clot location, n(%)	ICA			
	MCA-M1			
	Tandem occlusions			
Transferred to EVT capable center, n(%)				
IV thrombolysis administered, n(%)	total			
	IV Alteplase, n(%)			

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		Total	Allocated to endovascular thrombectomy + medical management (EVT arm)	Allocated to medical management only (MM arm)
	IV Tenecteplase, n(%)			
NIH Stroke Scale score at presentation to EVT capable center, median [IQR]				
General Anesthesia, n(%)				
Time parameters				
Time from last known well to randomization (minutes), Median [IQR]				
Time from arrival to CT acquisition (minutes), median [IQR]				
Time from arrival to CTP acquisition (minutes), median [IQR]				
Time from arrival to arterial puncture (Minutes), median [IQR]				
Time from arterial puncture to reperfusion/end of the procedure (Minutes), median [IQR]				
Imaging parameters				

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	Total	Allocated to endovascular thrombectomy + medical management (EVT arm)	Allocated to medical management only (MM arm)
ASPECTS on Baseline CT, median [IQR]			
Perfusion (rCBF <30%) Lesion Volume (ml), median [IQR]			
Tissue Volume with Tmax >6s (ml), median [IQR]			
Tissue Volume with Tmax >10s (ml), median [IQR]			

NIH – National Institutes of Health, ASPECTS – Alberta Stroke Program Early CT Score, EVT – Endovascular Thrombectomy, IQR – interquartile range, ml – milliliters

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Table 2. Clinical outcomes of intention-to-treat population stratified based on allocation of treatment.

		Allocated to endovascular thrombectomy + medical management (EVT arm)	Allocated to medical management only (MM arm)	Effect size with 95% confidence interval
		N=xx	N=xx	
Modified Rankin Scale score at 90-day follow-up, median [IQR]				
Functionally independent (mRS 0-2) at 90-day follow-up, n(%)				
Independent ambulation (mRS 0-3) at 90-day follow-up, n(%)				
Symptomatic ICH within 24 hours*, n(%)				
Neurological worsening within 24 hours, n(%)#				
All-cause mortality within 90-day follow-up, n(%)				
Procedural complications, n(%)	Total, n(%)			

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		Allocated to endovascular thrombectomy + medical management (EVT arm)	Allocated to medical management only (MM arm)	Effect size with 95% confidence interval
	Surgical site hematoma, n(%)			
	Surgical site infection, n(%)			
	Vascular injury due to EVT procedure, n(%)[+]			
Successful reperfusion (mTICI ≥2b), n(%)				
Discharge disposition, n(%)	Home, n(%)			
	Acute care facility, n(%)			
	Inpatient rehabilitation facility, n(%)			
	Skilled nursing facility, n(%)			
	Hospice care, n(%)			
	In-hospital death, n(%)			
Early Neurological Improvement, n(%)[$]				
Quality of life score				

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	Allocated to endovascular thrombectomy + medical management (EVT arm)	Allocated to medical management only (MM arm)	Effect size with 95% confidence interval
(NeuroQoL), median (IQR)			

*Defined as Parenchymal hemorrhage type 2 or remote parenchymal hemorrhage associated with an increase of 4 or more points in NIHSS score at 24 hour follow-up window (Based on SITS-MOST criteria after core lab adjudication of hemorrhagic transformation)

#Defined as an increase of 4 or more points in NIHSS score at 24 hour follow-up window compared to NIHSS at presentation to EVT capable center

+Includes xx patients with peripheral and intracranial dissection, xx patients with vascular perforation, and xx patients with other vascular complications

$Defined as improvement of at least 8 points in NIHSS compared to NIHSS at presentation to EVT capable center or NIHSS 0-1 at 24 hour follow-up window

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Figure 1: Distribution of modified Rankin Scale score at 90-day follow-up stratified based on treatment assignment in intention-to-treat population.

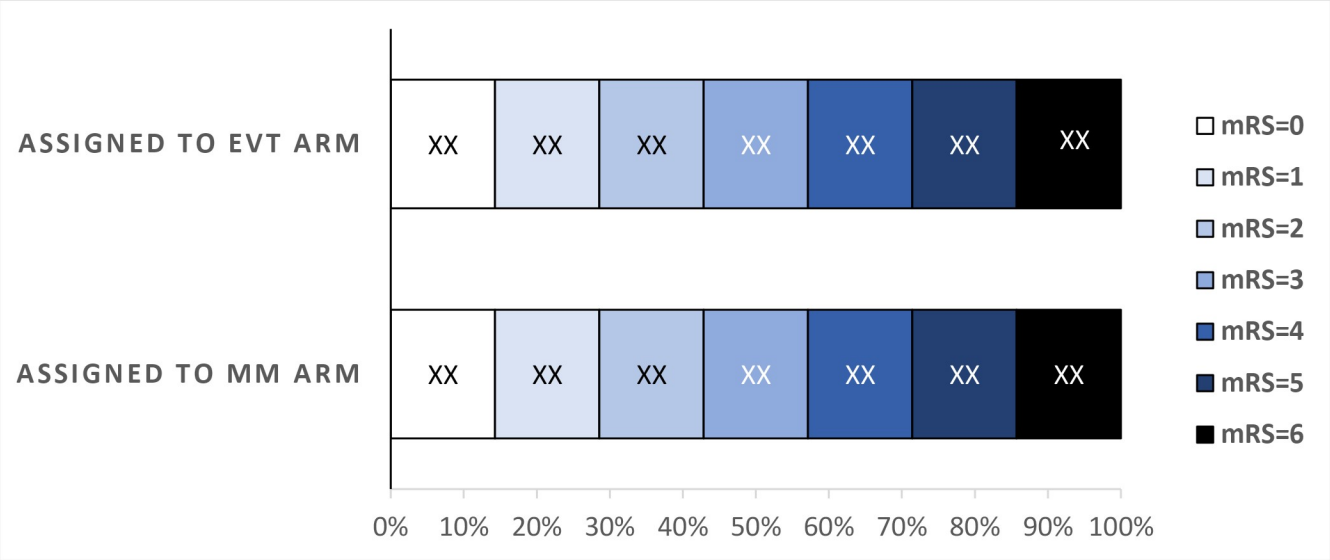

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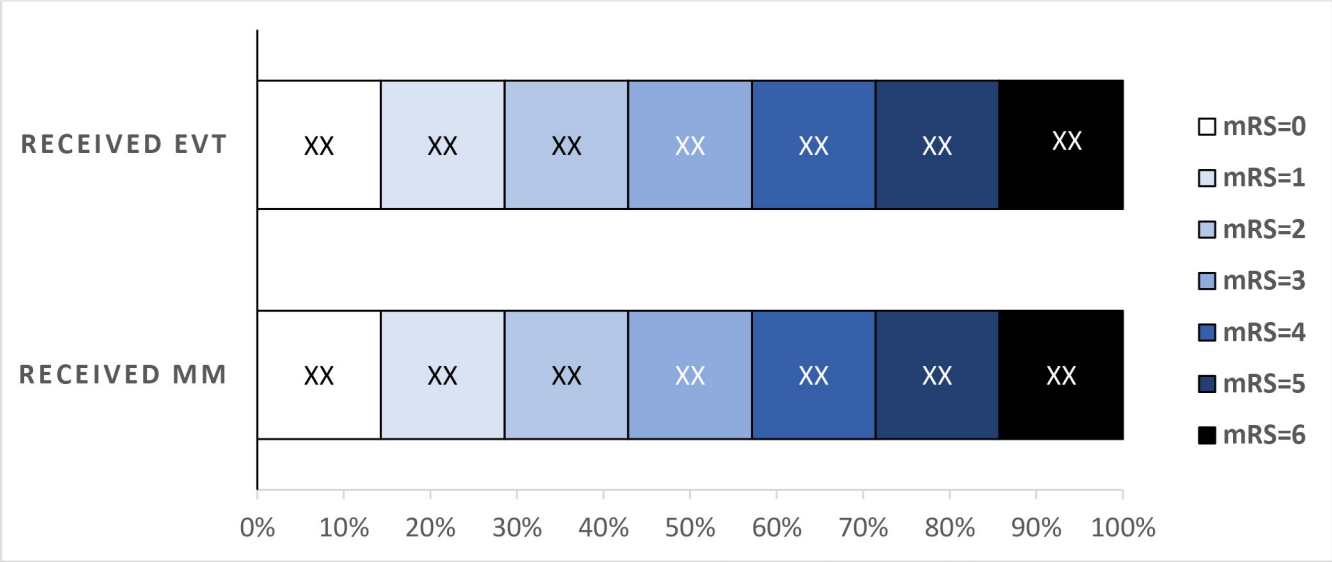
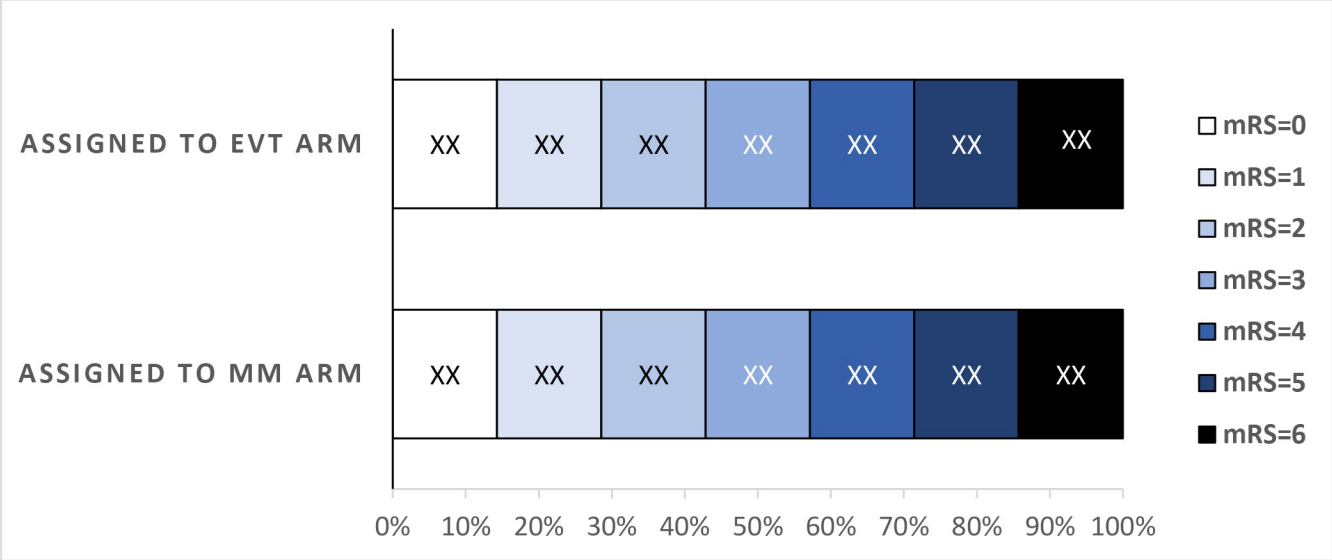
Figure 2: Distribution of modified Rankin Scale score at 90-day follow-up in pre-defined alternate populations.

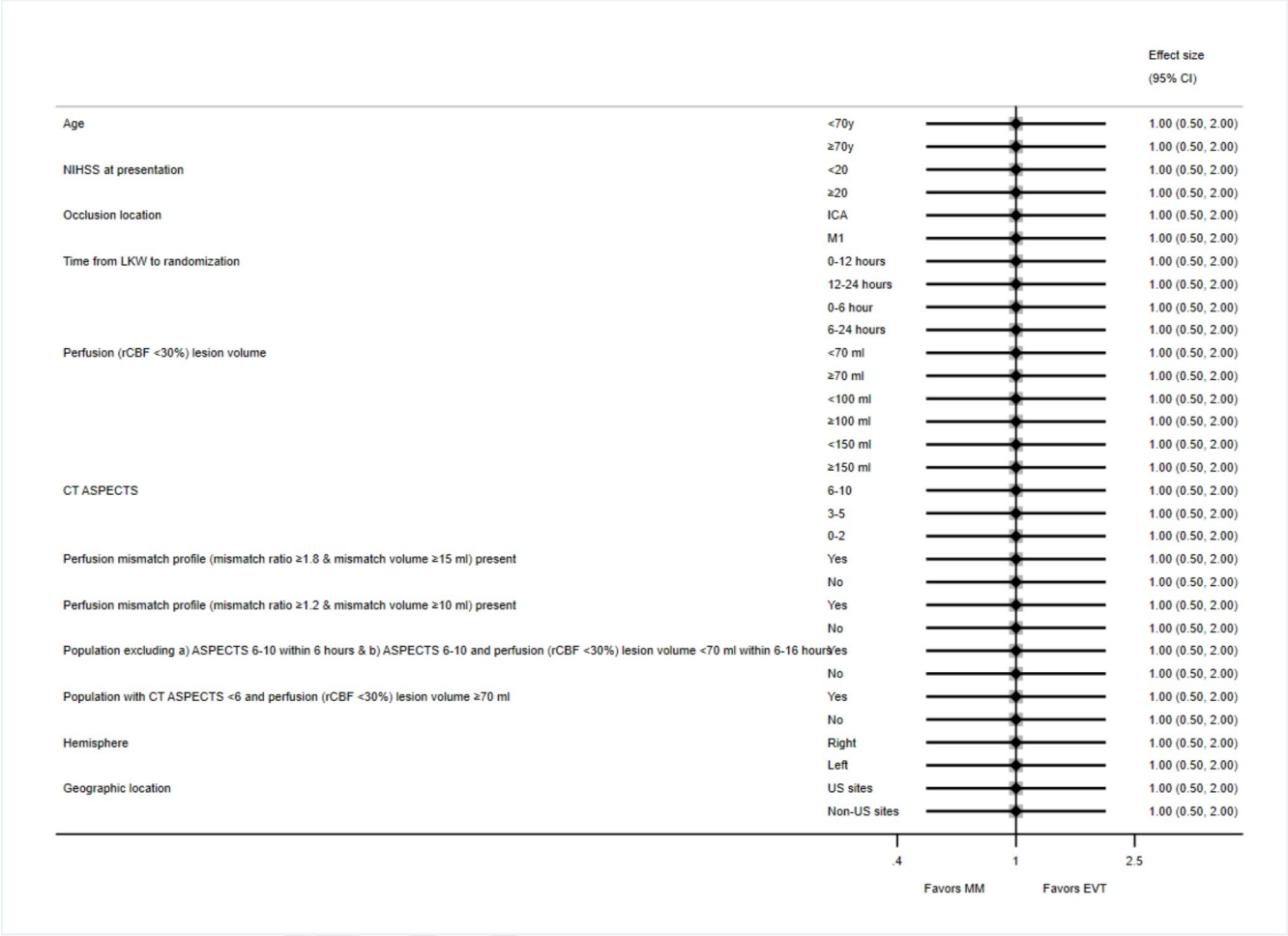


A) As-treated population



B) Per-protocol population

Figure 3. Forest plot illustration of EVT effect size across pre-specified subgroups in intention-to-treat population.



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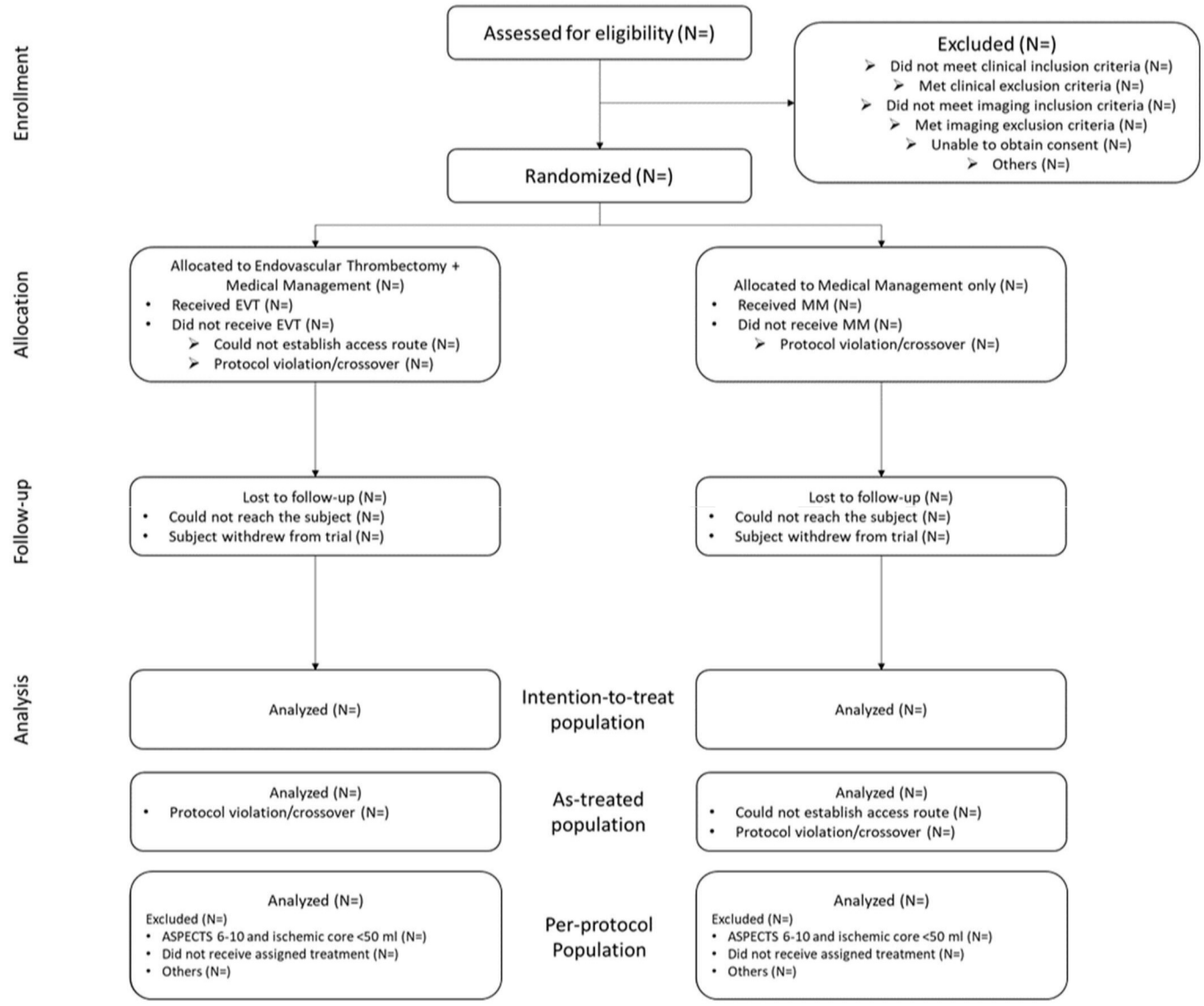
Supplemental Figure 1. CONSORT diagram



*Not mutually exclusive

Appendix A. Details of the minimization randomization algorithm and its implementation

Part I: Specification

This section provides a general framework for a study with the implementation of the minimization algorithm proposed by Pocock and Simon (1975).

Assumptions and Notations:

Assume there are K treatments denoted as $T_1, T_2, \dots, T_K$ with the randomization ratio as 1:1:…:1.

Assume there are S variables to be balanced in the minimization algorithm denoted as $X_1, X_2, \dots, X_S$.

For the variable X_f, suppose there are Q_f levels and denote the level of X_f as $Z_f \in \{1, \dots, Q_f\}$.

Algorithm:

When a new patient enters to the study, denote the observed level of the variable X_f for this patient as l_f, $f = 1, \dots, S$.

Step 1: Calculate the number of existing patients who are at the same level of the variable X_f as this patient in treatment k, denoted as $N_k(Z_f = l_f)$, $k = 1, \dots, K$. So we can obtain $N_1(Z_f = l_f)$, $N_2(Z_f = l_f)$,…, $N_K(Z_f = l_f)$.

Step 2: Regarding to the variable X_f, if we assign this patient to treatment k, we should have $N_k(Z_f = l_f)$+1 patients at treatment k who have the same level of the variable X_f as this patient. We obtain the amount of imbalance which would arise by assigning the patient to treatment k as

$$I_k(Z_f = l_f) = D(N_1(Z_f = l_f),\dots, N_k(Z_f = l_f) + 1,\dots, N_K(Z_f = l_f)).$$

Here, $D(\boldsymbol{a})$ is a function to measure the amount of variation for any set of non-negative integers $\boldsymbol{a}$. Then we can calculate the overall imbalance for all S variables arising when assigning the patient to treatment k by taking a weighted average, that is,

$$G_k = \sum_{f=1}^{S} w_f I_k(Z_f = l_f), k = 1, \dots, K,$$

where w_f denotes the weight for the variable X_f.

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Step 3: Find k^*=$\underset{k}{\operatorname{argmin}}\, G_k$ and assign the patient to treatment $k^*$with the highest probability among all treatments.

If we consider variance function for D, according to Freedman and White (1978), the total imbalance arising when assigning the patient to treatment k can be simplified as follow

$$G_k = 2(\sum_{f=1}^{S} w_f N_k(Z_f = l_f))/(K-1) + constant,\ k = 1, \dots, K.$$

Therefore, k^*=$\underset{k}{\operatorname{argmin}} \sum_{f=1}^{S} w_f N_k(Z_f = l_f)$

Part II: Implementation in SELECT2

Two treatment groups: T_1 and T_2 with 1:1 ratio.

	Level	Weight
Age	**<60, 60-69, 70-79, 80-85 years old**	**3**
Presentation NIHSS	**6-9, 10-15, 16-20, >20**	**4**
Clot location	**M1, ICA**	**3**
Treatment time window from onset to randomization time	**0-8, 9-16, 17-24 hours**	**3**
CTP/MRP core volume	**<50cc, 50-100cc, 101-150, 151-200, >200**	**4**
Target mismatch profiles	**0, 1**	**2**
Hemisphere	**right, left**	**2**
Center		**1**
Physician ASPECTS	**3-4, 5-6, 7-8, 9-10**	**3**

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For the first patient to be enrolled, we assign this patient to one of the treatments using equal randomization. Then before we assign next patients, we will calculate G_1 and G_2, if $G_1 > G_2$, then assign next patient to T_2 with a probability p. Here, we set $p = 0.9$; if $G_1 = G_2$, then assign patient to T_2 with probability 0.5; if $G_1 < G_2$, then assign patient to T_2 with probability=1-$p = 0.1$.

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Appendix B. Rules for defining efficacy and futility boundaries at interim and final analyses

At the lth interim analysis, let W_i^l be the sum of the ranks of Rankin scores from the endovascular group in the patient subgroup Π_i, $j = 1, \ldots, J$. In our trial, we defined J =6 groups with Π_J denotes the entire patient population and $\Pi_1 \ldots, \Pi_5$ denote the subgroups of G1A, G1A+G1B, G1A+G1B +G2A, G1A+G1B +G2A +G2B, G1A+G1B +G2A +G2B +G3A, respectively.

The standardized Wilcoxon statistic for patient subgroup Π_i is

$$Z_i^l = \left\{ W_i^l - \frac{n_i^l\left(n_i^l + n_{0i}^l + 1\right)}{2} \right\} / \left\{ \frac{n_i^l n_{0i}^l\left(n_i^l + n_{0i}^l + 1\right)}{12} \right\}^{1/2},$$

where, n_i^l and n_{0i}^l denote the numbers of patients enrolled into the endovascular group and medical management group at the lth interim analysis, respectively.

At the lth interim analysis, we will terminate the trial and claims efficacy for the endovascular group if the following efficacy boundary is crossed, that is,

$$Z_J^l \geq b.$$

Otherwise, we will proceed to the subgroup selection if the following futility boundary is crossed, that is,

$$\widetilde{Z_J^l} \leq \tilde{b},$$

Where,

$$\widetilde{Z_J^l} = \left\{ W_i^l - \frac{n_i^l\left(n_i^l + n_{0i}^l + 1\right)}{2} - n_i^l n_{0i}^l \theta \right\} / \left\{ \frac{n_i^l n_{0i}^l\left(n_i^l + n_{0i}^l + 1\right)}{12} \right\}^{1/2}.$$

When the above criteria is satisfied, a subgroup $\hat{I}$ with the largest value of Z_i^l for $i \neq J$ is chosen. The future enrollment of the trial will include patients of this subgroup only, while the maximum total sample size is still N=560.

Similar to testing the efficacy and futility in the entire population, at stage l', we will terminate the trial and claim efficacy for the endovascular group in this selected subgroup if the following efficacy boundary is crossed

$$Z_{\hat{I}}^{l'} \geq b.$$

We may also terminate for futility if

$$\widetilde{Z_{\hat{I}}^{l'}} \leq \tilde{b}.$$

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If neither the efficacy nor futility boundary for the entire population ever occurs, the trial proceeds to the final stage and we terminate the trial and claim efficacy for the endovascular group if

$$Z_J^{l=3} \geq c.$$

Under our design setting (maximum sample size N=560 and conducting interim analysis at N=200 and N=380 with type I error at 0.025 and power at 0.9), we obtain $b = 2.604, \tilde{b} = -1.897, c = 2.772$ for our efficacy and futility boundaries. We assume θ to be the projected treatment effect given the maximum sample size for testing the hypothesis: $H_0: P(Y \leq X) = \frac{1}{2}, H_A = P(Y \leq X) = \frac{1}{2} + \theta$ using the fixed-sample-size test and is calculated as $\frac{(Z_{1-\alpha}+Z_{1-\beta})}{\sqrt{3N}} = 0.0791$. Here, Y denotes the Rankin score in the medical management group and X denotes the Rankin score in the endovascular group. The rationale of this set-up is detailed in section 3.1 of Lai et al (2014).

References:

Lai T, Lavori P, Liao O. Adaptive choice of patient subgroup for comparing two treatments. *Contemporary Clinical Trials*. 2014;39(2):191-200. doi:10.1016/j.cct.2014.09.001

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